Identification of Undocumented Meat Samples Seized at Heathrow Airport

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Scenario

A suitcase containing several packages of unidentified frozen meat has been intercepted at Heathrow Airport by UK Border Force officers during routine inspections. The shipment had no import documentation, raising concerns that the meat may originate from protected or endangered species.

Given the serious implications for public health, food safety, and wildlife conservation, the case has been referred to the Animal and Plant Health Agency (APHA). As a bioinformatician assisting with the inquiry, you have been tasked with determining the species of origin for each meat sample. DNA barcoding has already been performed in the laboratory, and you have now received the resulting FASTAQ sequence files for analysis.

Abstract

Introduction

Rarity status attributed to a species often increases its price and demand, with critically endangered species fetching the highest prices [1, 2]. The fallout of this is a positive feedback loop between rarity and demand which may drive an endangered species towards extinction and threaten whole ecosystems [3]. Species-level identification of undocumented meat samples is key in public health for validating the meat being delivered to consumers, but may also uncover networks carrying out illegal imports of endangered species, thus aiding conservation efforts by providing routes to intervene in the trade of these species.

If the morphology of a sample is compromised, DNA barcoding - comparing genetic sequences in a sample against known references - can aid in resolving the species of the sample [4]. Mitochondrial DNA (mtDNA) genes are commonly used for DNA barcoding because they occur in high copy numbers relative to nuclear DNA and typically lack recombination, keeping within-species diversity low which enables species-level identification [5]. Cytochrome c oxidase I (COI), found within mtDNA, is the primary barcode marker in the animal kingdom, and reference sequences for COI are abundant in databases such as BOLD and NCBI, permitting barcoding against a diverse set of organisms [6, 7].

This analysis aims to resolve the species of four meat samples intercepted by UK Border Force officers at Heathrow Airport. The sequenced samples provide the starting point of this analysis pipeline.

Methods

Pipeline Overview

The pipeline consists of thirteen scripts which take the provided sequence data (`samples.zip`) through to the visualisation of a phylogenetic tree for sample identification. The minimal input design of the pipeline and provided `environment.yml` file maximises reproducibility and minimises incorporating bias or human error into the analysis. The four key pipeline stages are described here.

QC and Pre-Processing

FASTQ sequence files were first extracted from `samples.zip` before undergoing quality control to enforce a consistent FASTQ structure - including headers and line counts. Every part file (1-3) belonging to the four samples (A-D) were then concatenated into multi-read FASTQ files by sample. `seqtk` [8] was used to convert each to FASTA while masking any bases with quality scores < 20 to 'N', ensuring bases were called at a confidence $\geq 99\%$ during sequencing to reduce the chances of false positives and increase confidence in downstream BLAST results.

BLAST and FASTA Concatenation

BLAST [9] searches were carried out using the FASTA files as query sequences against the NCBI Nucleotide database using `blastn` [10] with the results saved to tsv files. BLAST results displayed homology between parts across samples - for this reason parts were treated as homologous loci and hits were split back into tsv files by sample part to enable finer-tuned filtering and to create per-part alignments downstream. These results were then sorted by biological relevance (Fig. 1) and collapsed by species with `taxonkit` [11] staxid values before saving the top four results for each sample and part.

```
sort -k12,12g -k13,13gr -k5,5gr 5_blast_filtering/"$part_name"_blast.tsv \
| awk -F'\t' '!seen[$3]++' > top_hits.tsv
```

Figure 1: Bash line to sort BLAST hits by ascending e-value (12), descending bitscore (13), and descending length (5), saving only sorted-unique staxid hits (3).

`efetch` [12] was used to download the FASTA sequences for each BLAST hit accession, and the `sstart` and `ssend` values were used to trim these sequences to match the query region using `seqtk`, reverse complementing if an opposite strand is matched in BLAST. COI sequences of three Selachii species were retrieved using `efetch` to be used as an outgroup before all FASTA sequences - query, BLAST hits, and outgroups - were concatenated by part into three complete FASTA files (parts 1-3) using `seqkit` [13].

Translation and Alignment

Translation of the FASTA sequences using the vertebrate mitochondrial code was carried out using Biopython's [14] `.translate()` method and `table=2`. To counteract potential frame shift errors during translation, ORFs were identified by splitting each translated sequence by

STOP (*) codons, and the longest sequence containing no STOPs saved as the protein sequence for each read.

MUSCLE5 [15] was used to carry out multiple sequence alignments on the three part-protein files, and `trimAl` [16] trimmed the alignment files to reduce post-translational artefacts while retaining internal sites for a cleaner and more stable tree build downstream. To build a single tree from the three trimmed alignments, a supermatrix alignment file was produced using `catfasta2phyml` [17]. This produces a concatenated supermatrix alignment along with partitions for each part to allow for different mutation rates per loci within a single alignment upon tree building.

Tree Building and Viewing

The final tree file was built using `iqtree3` [18] from the supermatrix alignment file and provided partition file. The Python library `ete3` [19] was used to visually display the phylogenetic tree as well as rooting to the outgroup Selachii clade.

Results

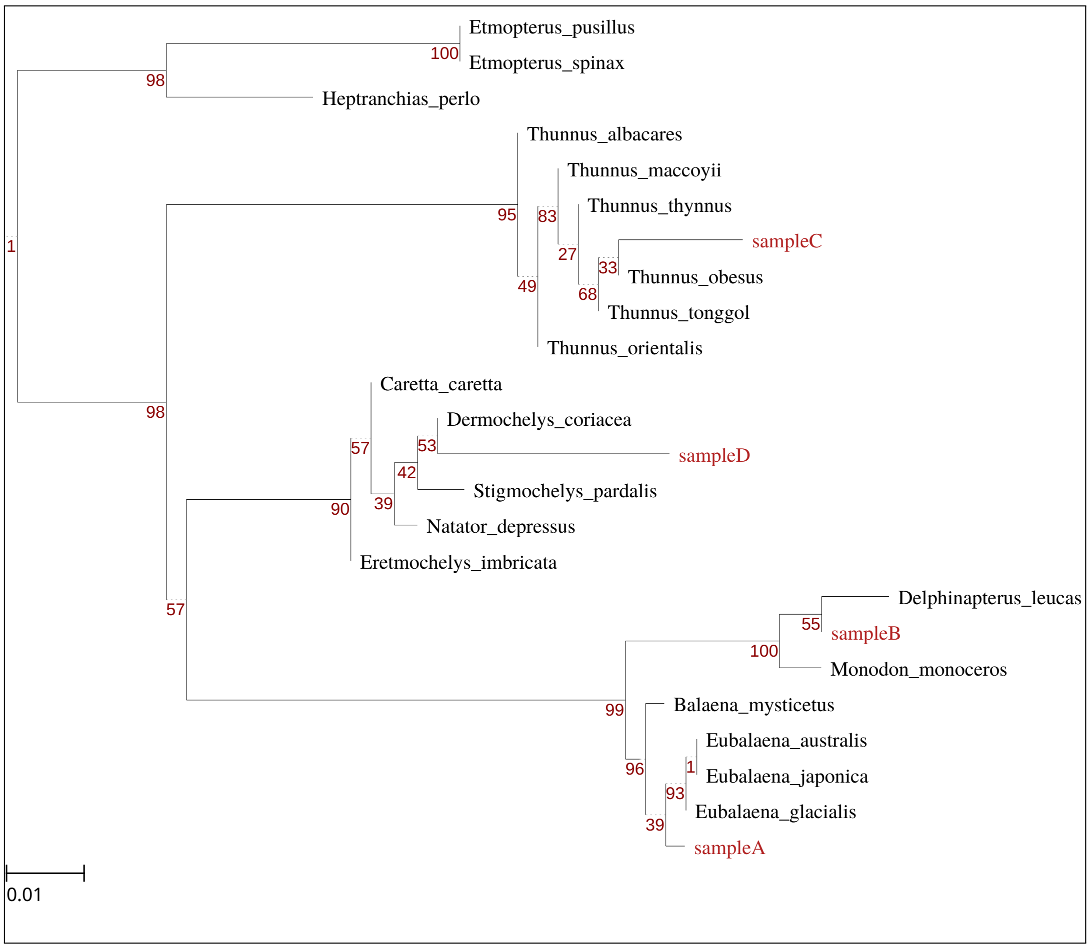

Figure 2: Maximum-likelihood phylogenetic tree with placement of unknown meat samples and top BLAST hits. Created with `ete3` [19].

Discussion

Where cyt b and D-Loop sequences do exist, very few of these have been generated as part of forensic studies [23], with the vast majority submitted to databases as part of phylogenetic, molecular ecological and conservation studies [24], [25], [26]. Such sequence data can be generated and accepted into databases without any need for standard protocols or quality control, raising doubts over their suitability for forensic application [5]

However, unlike BOLD, GenBank does not store sequence chromatograms, collection metadata or photographs [6]. BOLD is a curation tool that also stores sequences, while GenBank is just a sequence repository. [20]

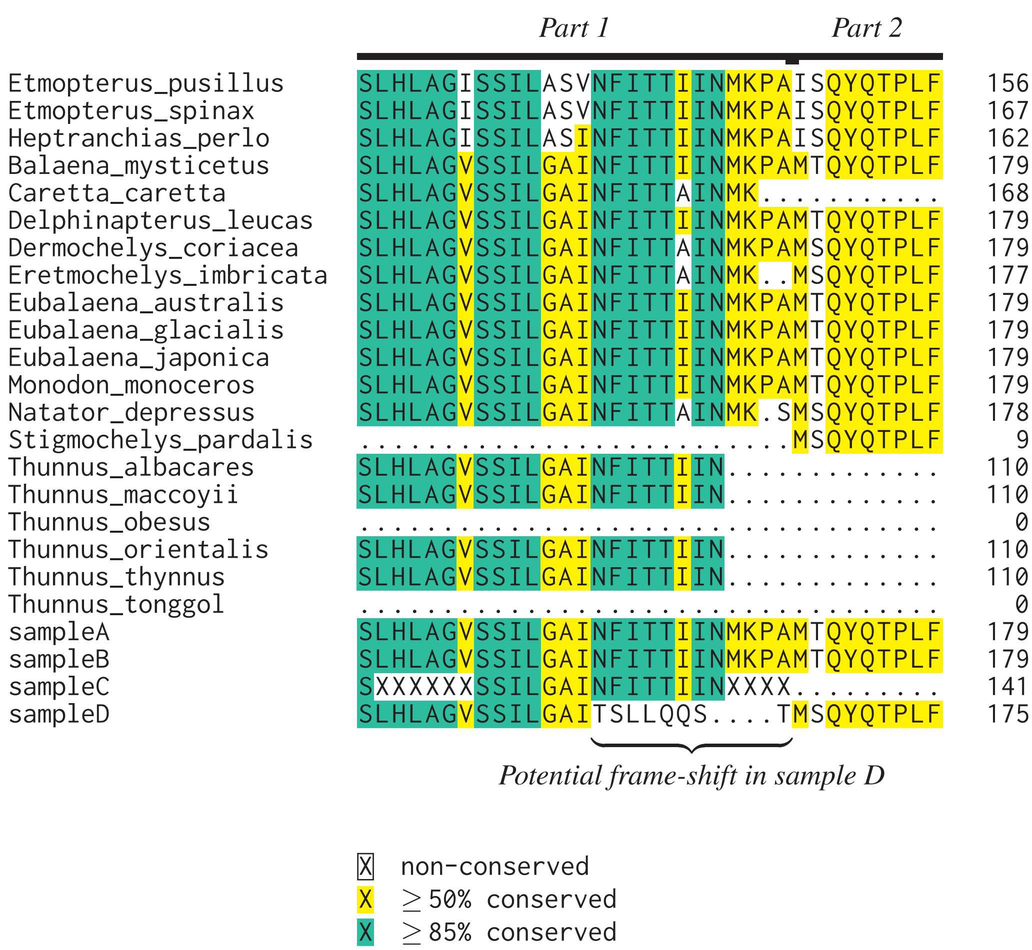

Figure 3: Window of supermatrix alignment with potential frame-shift error in `sampleD` near the highly-conserved `NFITTIIN` region.

```
Dermochelys_c    441 CACCTAGCTGGTGTTTCATCAATTTTAGGAGCTATTAACTTCATTACTAC    490
                     |||||||||||||||||||||||||||||||||||| |||||||||||||
sampleD_part1    451 CACCTAGCTGGTGTTTCATCAATTTTAGGAGCTATT-ACTTCATTACTAC    499
```

Figure 4: Needleman-Wunsch global pairwise alignment at the tail end of `sampleD_part1` against closest reference sequence. Alignment gap indicates a possible base miss in `sampleD_part1` during sequencing.

References

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Supplementary Data

Supplementary data and all scripts can be found within the GitHub repository for this project: https://github.com/archiebenn/BIOLM0051_analysis.git

Appendix: Exploratory Figures in Suspected Sequencing Error

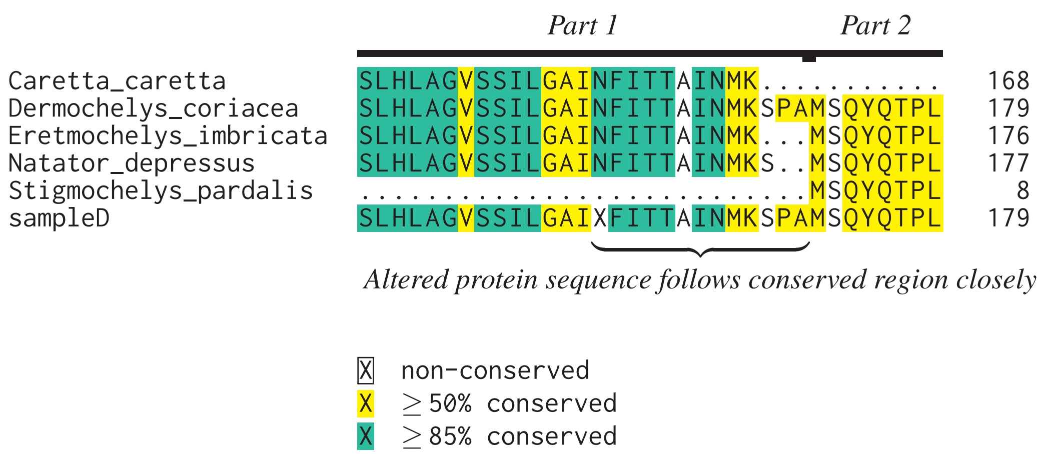

Figure 5: Subset of supermatrix alignment with `sampleD` and innner clade after inserting one ambiguous base (N) to the nucleotide sequence of `sampleD_part1`. This base resolves the apparent frame-shift, resulting in a protein sequence which aligns much more closely to highly-conserved `NFITTIIN` region compared to in Figure 3.

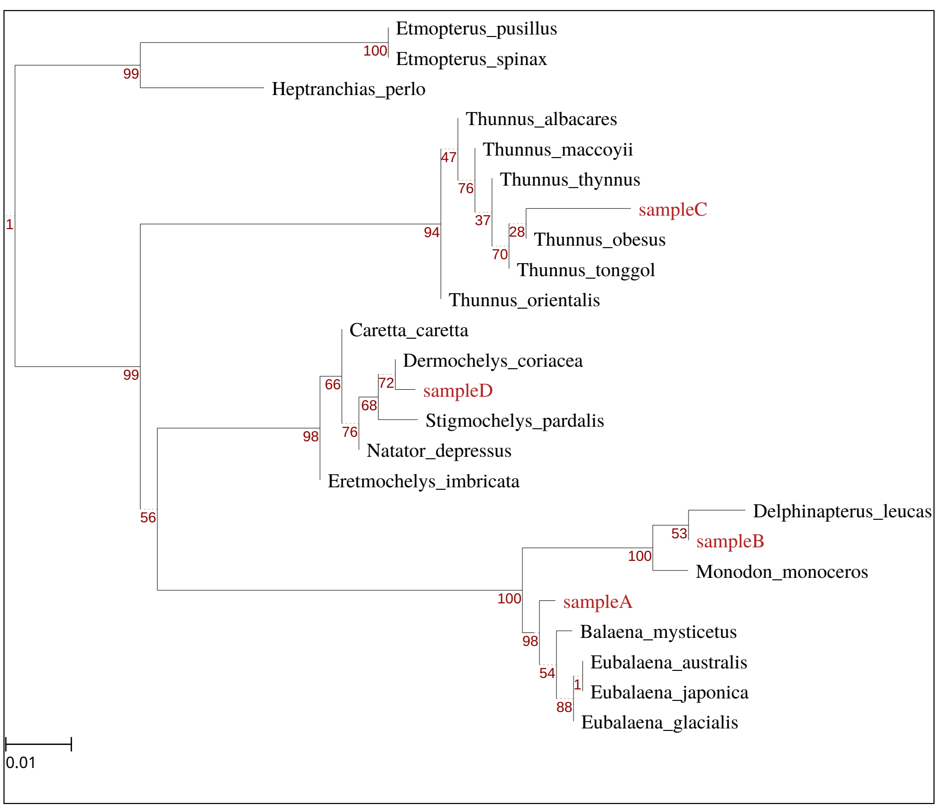

Figure 6: Maximum-likelihood phylogenetic tree after manually corrected alignment in which one ambiguous base (N) was inserted into `sampleD_part1`. This tree is shown to demonstrate the impact of sequencing error and downstream frame-shift on branch lengths compared to Figure 2. Created with `ete3` [19].